

Fathom



AI Systems Architecture Review

Advisor stock briefings grounded in SEC filings — quote, filings, verified-citation AI briefing, grounded Q&A

Executive summary

What it is

- A one-page briefing for a wealth-management advisor: current quote with context, the latest 10-K and 10-Qs with EDGAR links, an AI briefing in six sections, and follow-up Q&A.
- Every AI claim carries a verbatim quote that the application re-checks against the filing before it is shown.

What it proves

- Grounded generation can be made auditable: claims are verified, advice language is blocked, and every model call leaves a hash-only audit record.
- The same pipeline runs on 20 curated tickers (fixtures) or on any US-listed ticker from free, keyless sources.

Why it is enterprise-ready

- Built through a gated, evidence-producing lifecycle: 25 task packs, 25 independent verifications, 14 security reviews, 43 hash-chained ledger entries, CI gate on every push.
- Clear trust boundaries, a threat model, a runbook, an ORR and a change record.

Ask of the audience: agree the pilot scope (full EDGAR universe, SSO, tamper-evident audit, advisor eval set) and the data-source policy for live quotes.

The problem and the users

The advisor's problem

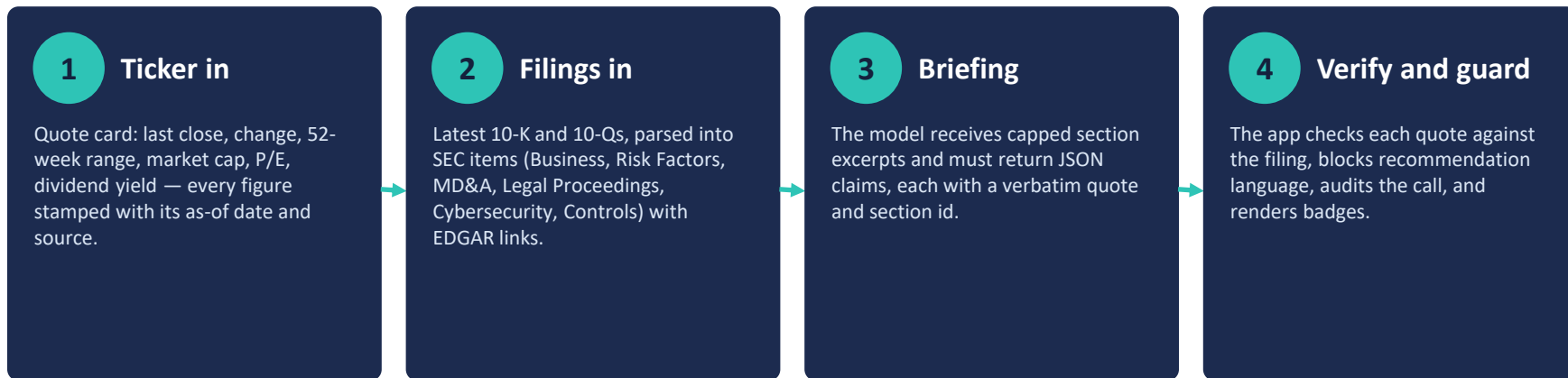
A client calls about a name the advisor does not follow. Getting current takes 30–60 minutes across a quote screen and two or three SEC filings. A general-purpose chatbot is faster but cannot be quoted to a client: it paraphrases, it may recommend, and it leaves no record.

The firm's constraints

Wealth management is a supervised activity. Anything the tool says may be repeated to a client, so it must describe and never advise, cite sources an advisor can open, and keep a record of what the AI produced from which inputs.

Actor	Needs	Frequency
Advisor (primary)	One page per ticker: quote context, recent filings, cited briefing, follow-up questions; no advice language	Several times a day
Compliance / supervision	Audit trail: what was produced, from which filings, by which model; evidence the guards work	On review
Firm platform team	Runs on the firm's AI gateway; no keys in code; offline fallback; one command to validate	Deploy / operate
Automation and AI assistants	The same capabilities over HTTP and MCP with identical contracts	Ad hoc

Solution at a glance



Surfaces

Streamlit advisor page · Typer CLI · FastAPI (JSON envelope) · MCP server (four tools). All four call the same package functions; none computes anything itself.

Two data modes, one schema

Fixtures: 20 large caps, 97 filings, committed. Live: any US-listed ticker from SEC EDGAR + Yahoo/Stooq, materialized per ticker into the same four tables. Switch with one environment variable.

System context



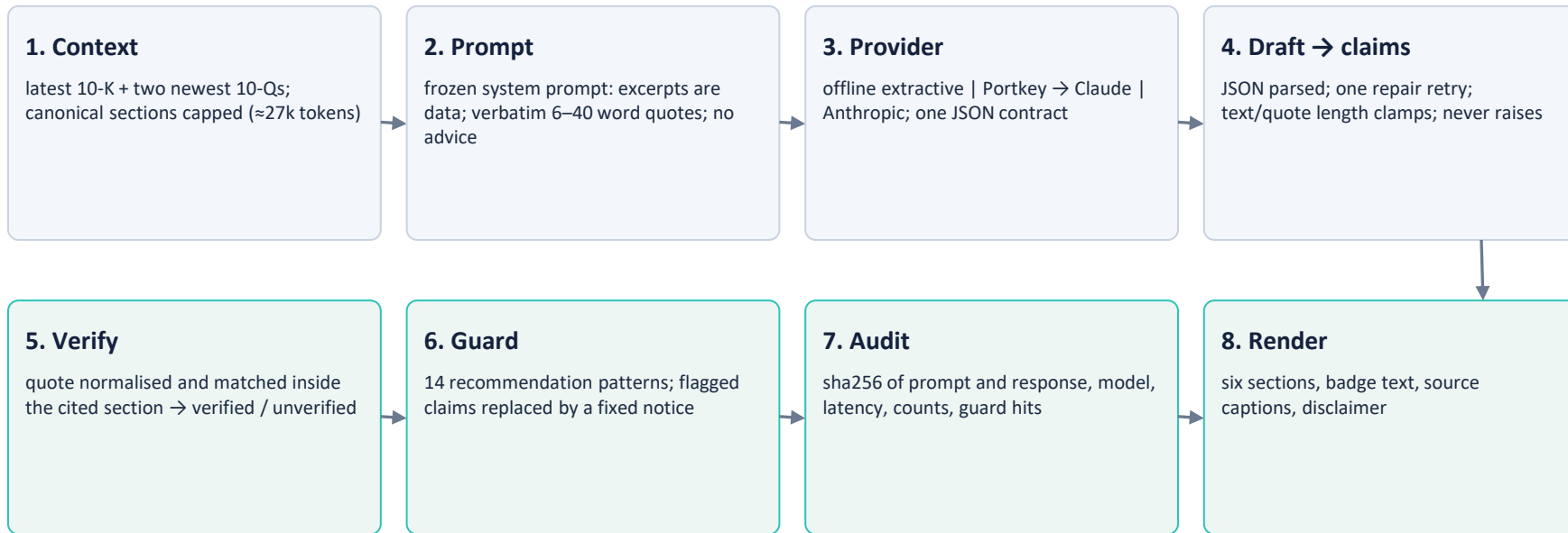
Build-time only: Hugging Face datasets → fixtures via scripts/fetch_data.py. Live sources are called only when FATHOM_DATA_SOURCE=live; the gate and CI never touch the network.

Component architecture



Dependency direction is left to right only. Surfaces never import pandas or regex; providers never see settings beyond their own keys; the data layer knows nothing about models.

The briefing pipeline



Failure paths are first-class: invalid JSON → one repair → CONTRACT_INVALID; gateway down → PROVIDER_HTTP with a reason and no body; unverifiable quote → shown and flagged, never hidden.

Verify, don't trust: the claim contract

What the model must return (one claim)

```
{
  "text": "Net sales increased 5% year over year",
  "accession": "0000320193-26-000013",
  "section_id": "10-Q:I.2",
  "quote": "Total net sales increased 5% or $4.5 billion
during the second quarter of 2026"
}
```

What the application does with it

- normalise whitespace, case and quote glyphs on both sides
- require 6–60 words and a substring match inside the cited section's full text
- set verified = true/false; scrub advice language; count; audit

**verified**

quote found in the cited section

**unverified**

shown, flagged, counted — never hidden

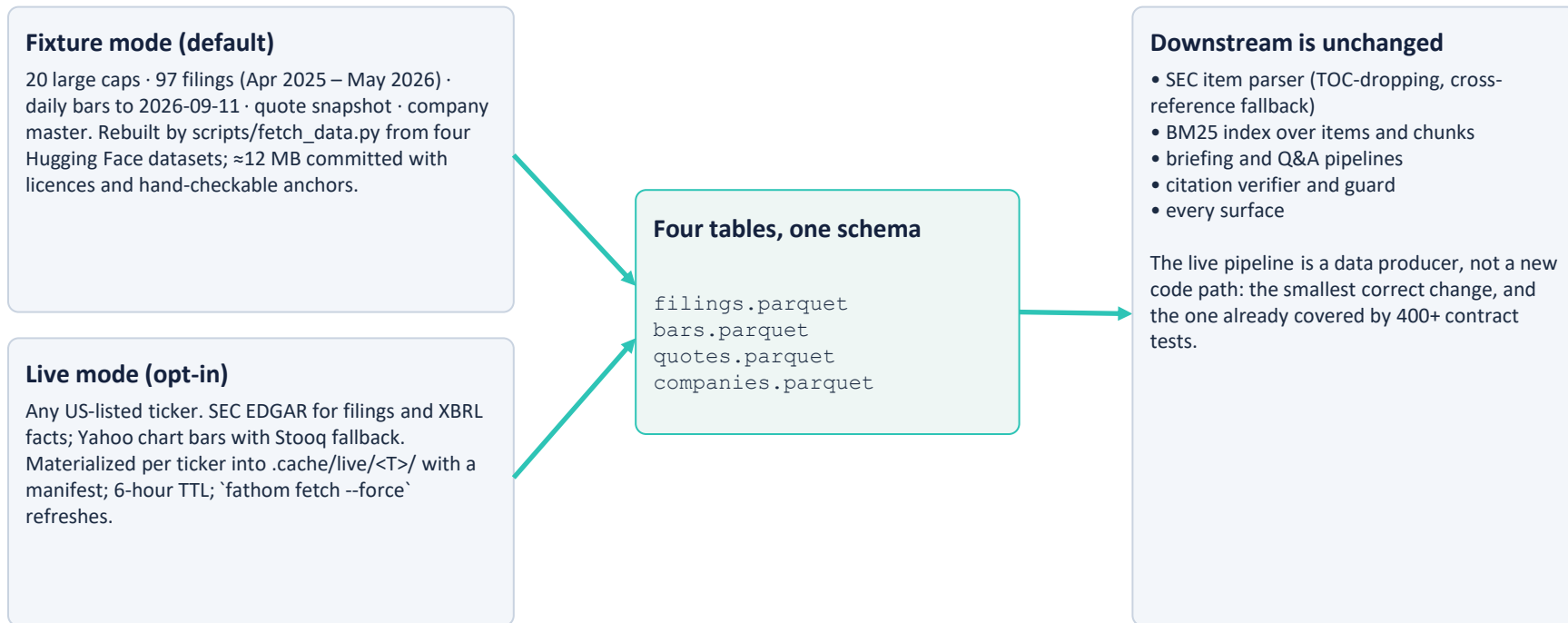
**removed**

recommendation-style language replaced by a fixed notice

Why this matters to compliance

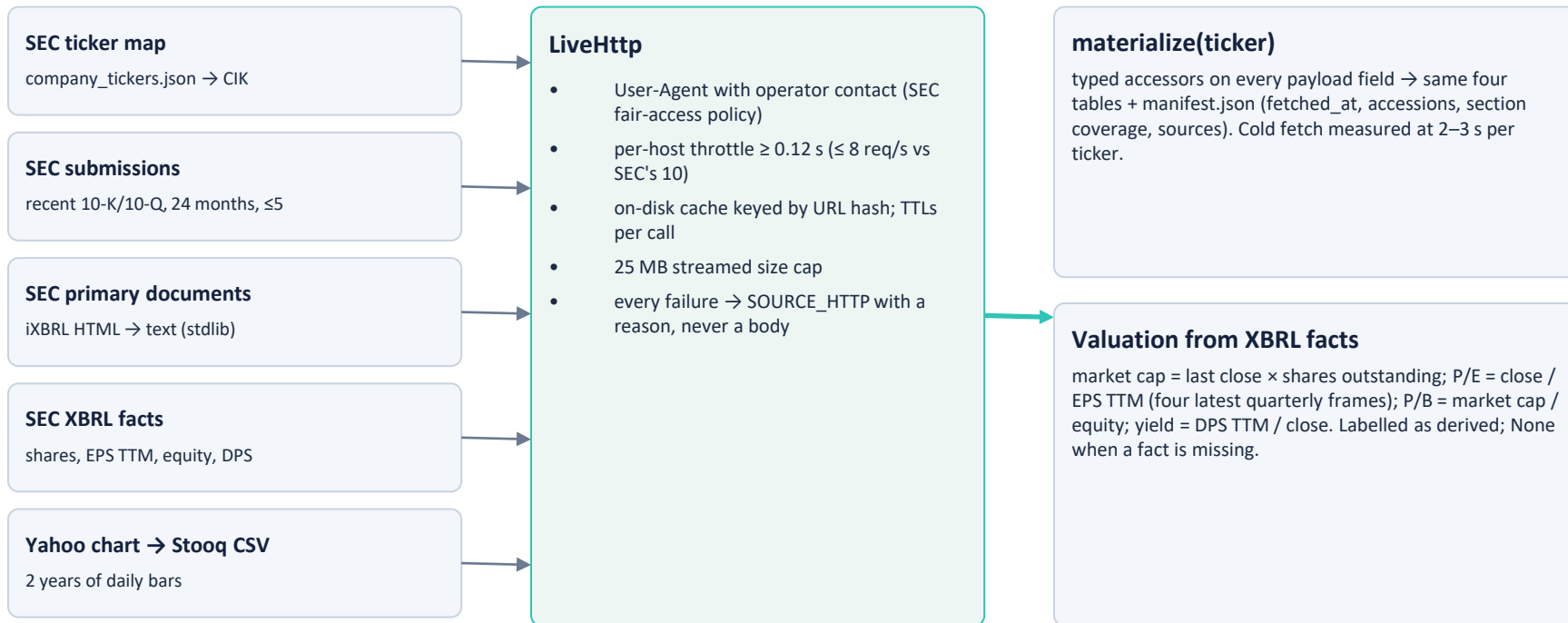
The badge is the control, and the eval suite reports the verified share as a headline KPI. Offline mode is verified by construction (its quotes are the sentences themselves), so it proves the plumbing; live-mode quality is measured from the audit log, not asserted.

Data architecture: two sources, one schema



Design rule: `data\_dir\_for(ticker, settings)` is the only routing decision; the ticker is shape-validated before any path or URL is built.

Live data pipeline (free, no API keys)



Tested offline with recorded fixtures and structural fuzzing; a network smoke test is opt-in (FATHOM_NETWORK_TESTS=1).

Design choices and why

Decision	Alternatives considered	Why this one
Verifiable claim contract (D-003)	Free-text summary; summary with section references	A briefing an advisor may repeat must be checkable; the badge becomes the control and the KPI
Lexical BM25 over SEC items, no vector DB (D-004)	Embeddings + FAISS/Chroma; rank_bm25	Five documents per ticker already structured into items; explainable to compliance; deterministic in CI; one function to swap later
Offline provider is a real provider (D-005)	Branch in the pipeline between extractive and LLM	One code path from prompt to verified claims; demo, CI and bench never need a key
Plain httpx, no vendor SDKs	OpenAI / Anthropic / Portkey SDKs	The verifier and error taxonomy are the product; three small clients are easier to audit than three SDK surfaces
Live mode materializes the fixture schema (D-013)	DataSource protocol refactor; keyed vendor APIs	Parser, retrieval, briefing and guard stay untouched and contract-tested; no keys, SEC is official
Typed payload accessors + structural fuzz (D-014)	Enumerated exception guards	Four security rounds showed enumerated guards leak; typed accessors and node-mutation fuzz closed the class
Streamlit + Typer + FastAPI + MCP	React front end; n8n canvas	Core functionality over polish; every surface is a thin renderer over one package

Major features

Quote card

last close and change, 52-week range, market cap, P/E, dividend yield, one-year chart; each with an as-of stamp and source

Filings table

latest 10-K and 10-Qs with fiscal period, accession and a link to the EDGAR index

AI briefing

six sections: business snapshot, latest results, risks, liquidity and capital, notable disclosures, talking points

Grounded Q&A

advisor question → BM25 over items and chunks → cited answer; "not found" when the filings do not cover it

Advice guard

input and output patterns for buy/sell/hold, price targets, over/underweight, valuation calls; fixed notice on hit

Audit log

one JSON line per call: hashes, model, latency, tokens, claim counts, guard hits; bodies only when opted in

Four surfaces

Streamlit page, CLI (brief/ask/quote/filings/fetch/probe/bench), HTTP API with a JSON envelope, MCP tools

Live data mode

any US-listed ticker from SEC EDGAR + Yahoo/Stooq; 'fathom fetch'; cache and TTL; same contracts

Eval bench in the gate

parser coverage, retrieval hit-rate, guard escapes, verified share, injection test → metrics card

The advisor page: quote card, chart and filings

Fathom

Apple Inc.

NasdaqGS · Technology · Consumer Electronics

<https://www.apple.com>

Last close	52-wk low	52-wk high
332.27	228.18	344.27
↑ +1.75%		
Market cap	P/E	Dividend yield
\$4.85 T	38.10	0.32%

Prices: AlphaDojo/dojo_stock_kline via data/bars.parquet as of 2026-09-11 · Snapshot: AlphaDojo/dojo_quote via data/quotes.parquet as of 2026-09-11 16:00 UTC



Offline fixture mode. Six metrics with a single source caption; the one-year close series; the filings table links to the EDGAR index for each accession.

The briefing: every claim cited, every badge earned

Business snapshot

- The Company designs, manufactures and markets smartphones, personal computers, tablets, wearables and accessories, and sells a variety of related services.  verified

Source: 10-K 2025-10-31 · Business · 0000320193-25-000079

- The Company's fiscal year is the 52- or 53-week period that ends on the last Saturday of September.  verified

Source: 10-K 2025-10-31 · Business · 0000320193-25-000079

- Is the Company's line of smartphones based on its iOS operating system.  verified

Source: 10-K 2025-10-31 · Business · 0000320193-25-000079

Latest results

- Unless otherwise stated, all information presented herein is based on the Company's fiscal calendar, and references to particular years, quarters, months or periods refer to the Company's fiscal years ended in September and the associated quarters, months and periods of those fiscal years.  verified

Source: 10-Q 2026-05-01 · Management's Discussion and Analysis · 0000320193-26-000013

- Securities and Exchange Commission ("SEC") and the condensed consolidated financial statements and accompanying notes included in Part I, Item 1 of this Form 10-Q.  verified

Source: 10-Q 2026-05-01 · Management's Discussion and Analysis · 0000320193-26-000013

- The Company periodically provides certain information for investors on its corporate website, www.apple.com, and its investor relations website, investor.apple.com.  verified

Source: 10-Q 2026-05-01 · Management's Discussion and Analysis · 0000320193-26-000013

- This includes press releases and other information about financial performance, information on

What you are looking at

Offline extractive provider: sentences lifted verbatim from the cited sections, so every badge is  verified by construction. Each claim carries a caption with form, filing date, section title and accession.

In live LLM mode

Claude Sonnet 4.5 via Portkey writes the claim text and must copy a 6–40 word quote; paraphrased quotes surface as  unverified. The status strip shows provider, model, claim counts and verified counts for the session.

Grounded Q&A and the advice guard

Answer

- Tariffs have been announced and further changes are expected to be made in the future, which may include additional sector-based tariffs or other measures.  verified

Source: 10-Q 2025-05-02 · Risk Factors · 0000320193-25-000057

- Tariffs have been announced and further changes could be made in the future, which may include additional sector-based tariffs or other measures.  verified

Source: 10-K 2025-10-31 · Risk Factors · 0000320193-25-000079

- Tariffs and other measures that are applied to the Company's products or their components can have a material adverse impact on the Company's business, results of operations and financial condition, including impacting the Company's supply chain, the availability of rare earths and other raw materials and components, pricing and gross margin.  verified

Source: 10-Q 2025-05-02 · Management's Discussion and Analysis · 0000320193-25-000057

Provider: offline · Model: extractive-v1 · claims 20 · verified 20

Cited answers

"What are the main risk factors?" → the top-ranked chunks, answered with verified quotes from Risk Factors.

Guarded questions

"Should I buy Apple stock?" never reaches the model: the input guard returns the fixed notice and the call is audited with provider = guard.

Live mode: a ticker outside the demo set

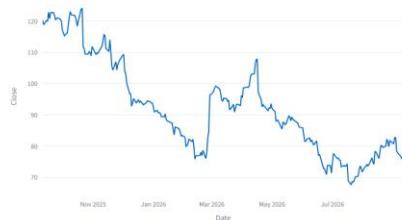
Fathom

NETFLIX INC

Nasdaq - Services-Video Tape Rental - Services-Video Tape Rental

Last close	52-wk low	52-wk high
77.40	65.08	124.86
↑ +1.83%		
Market cap	P/E	Dividend yield
\$322 B	8.98	0.00%

Prices: yahoo via .cache/live/NFLX/bars.parquet as of 2026-09-11 · Snapshot: SEC XBRL companyconcept (shares, EPS TTM, equity, DPS TTM) × yahoo close as of 2026-09-11 21:00 UTC



Form	Filing date	Period end	Accession	EDGAR link
10-Q	2026-07-17	2026-06-30	0001065280-26-000212	https://www.sec.gov/Archives/edgar/data/1065280/000106528026000212
10-Q	2026-04-17	2026-03-31	0001065280-26-000138	https://www.sec.gov/Archives/edgar/data/1065280/000106528026000138
10-K	2026-01-23	2025-12-31	0001065280-26-000034	https://www.sec.gov/Archives/edgar/data/1065280/000106528026000034
10-Q	2025-10-22	2025-09-30	0001065280-25-000406	https://www.sec.gov/Archives/edgar/data/1065280/000106528025000406
10-Q	2025-07-18	2025-06-30	0001065280-25-000323	https://www.sec.gov/Archives/edgar/data/1065280/000106528025000323

What happened

Sidebar switched to Live; ticker typed; Fetch. In about two seconds Fathom pulled five filings from SEC EDGAR, two years of Yahoo bars and four XBRL facts, and wrote the four tables plus a manifest.

Read the sources line

"Prices: yahoo via .cache/live/NFLX/bars.parquet as of 2026-09-11 · Snapshot: SEC XBRL companyconcept ... × yahoo close" — provenance is on the page, not in a footnote.

Live mode: briefing and answer on fresh filings

Business snapshot

- ("Netflix", the "Company", "registrant", "we", or "us") is one of the world's leading entertainment services offering TV series, films, games and live programming across a wide variety of genres and languages.  verified

Source: 10-K 2026-01-23 · Business · 0001065280-26-000034

- Members can play, pause and resume watching as much as they want, anytime, anywhere, and can change their plans at any time.  verified

Source: 10-K 2026-01-23 · Business · 0001065280-26-000034

- Our core strategy is to grow our business globally within the parameters of our operating margin target.  verified

Source: 10-K 2026-01-23 · Business · 0001065280-26-000034

Latest results

Risks

- If any of the following risks actually occur, our business, financial condition and results of operations could be harmed.  verified

Source: 10-K 2026-01-23 · Risk Factors · 0001065280-26-000034

- In that case, the trading price of our common stock could decline, and you could lose all or part of your investment.  verified

Source: 10-K 2026-01-23 · Risk Factors · 0001065280-26-000034

- If our efforts to attract and retain members are not successful, our business will be adversely affected.  verified

Source: 10-K 2026-01-23 · Risk Factors · 0001065280-26-000034

- We must continually add new members both to replace canceled memberships and to grow our business beyond our current membership base.  verified

Source: 10-K 2026-01-23 · Risk Factors · 0001065280-26-000034

Liquidity and capital

Notable disclosures

- We have an enterprise-wide information security program designed to identify, protect, detect and

Answer

- We also utilize third-party technology to help market our service, process payments, and otherwise manage the daily operations of our business.  verified

Source: 10-K 2026-01-23 · Risk Factors · 0001065280-26-000034

- Tax authorities are increasingly scrutinizing the tax positions of companies and we have tax audits pending in a number of jurisdictions.  verified

Source: 10-K 2026-01-23 · Risk Factors · 0001065280-26-000034

- In addition, the continued growth and development of the market for online commerce may lead to more stringent consumer protection laws, which may impose additional burdens on us.  verified

Source: 10-K 2026-01-23 · Risk Factors · 0001065280-26-000034

Provider: offline · Model: extractive-v1 · claims 12 · verified 12 · source: live · fetched 2026-09-12T20:34

Honest reading

Offline extractive provider on live text: the Risk Factors and Business sections read well; "Latest results" and "Liquidity" came back empty because Netflix's MD&A opens with tables and boilerplate that the extractive heuristics skip. The live LLM path composes those sections; the extractive path is the no-key fallback.

Security and compliance controls

Boundary	Threat	Control in the code	Evidence
B1 user input	advice elicitation, prompt override, free-text ticker	input + output guard; ticker shape check before any path/URL; API enum for source; question length bound	guard tests, T-020 security
B2 AI gateway	key leakage; malformed or spoofed responses	keys only from env by name; secrets scan in the gate; JSON contract with one repair; every httpx error mapped to a Fathom error	T-004/T-015 security
B3 filing text	prompt injection embedded in filings	excerpts declared as data; injected instructions cannot become verified claims; FR-018 test in the bench	bench injection KPI
B4 audit file	bodies, questions or keys in logs	hashes and counts only by default; never-log test with a capturing handler	T-005 security
B6 live sources	tampered or malformed payloads; rate limits; contact exposure	stdlib parsing; typed accessors; whole-body guards; streamed size cap; throttle; contact never committed	T-018 ... T-024 security

Residual risks are written down and owned: no authentication in the prototype (SSO in front for a pilot), local audit file not tamper-evident, unofficial price endpoints.

How it was built: a gated, evidence-producing lifecycle



25

task packs, each with acceptance criteria, scope, budget and risk class

25

independent verifications in a fresh context, in a git worktree pinned to the task's commit

14

security reviews; every HIGH closed by a fix pack and a fresh re-review

43

hash-chained evidence-ledger entries; chain verified at export

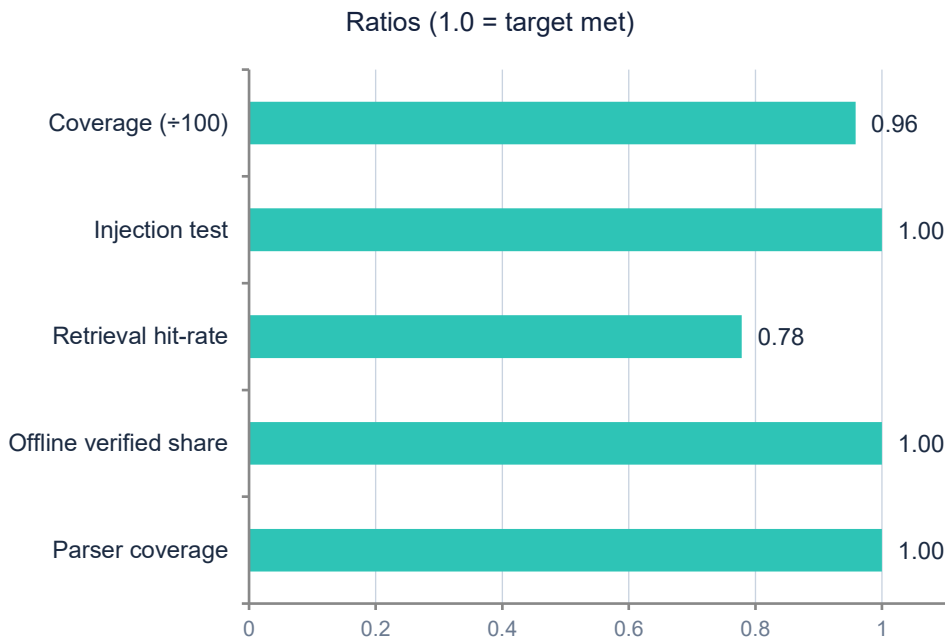
Separation of duties, by construction

Frontier model for design, threat model, contracts and orchestration; Sonnet-class agents for every implementation, verification and security review. The implementer never verifies; the verifier never edits; no agent writes an approver's name.

One command validates everything

ruff · mypy strict (host and Linux) · pytest with $\geq 80\%$ coverage · secrets scan · offline eval bench · drift checks. CI runs the same command on every push; the network is never touched.

Quality metrics from the gate



KPI	Value	Target
Tests / coverage	458 passed (2 network tests skipped offline) / 95.8 %	≥ 80 %
Guard escapes (30 adversarial phrases)	0	0
Guard false positives (12 benign)	0	info only
Offline briefing latency, median	33 ms	≤ 5 s
Live cold fetch (NFLX, COST)	≈ 2–3 s	≤ 30 s
Prompt size per briefing	≈ 27 k tokens	≤ 40 k
Live briefing latency	measured from the audit log on the day	≤ 90 s

Retrieval hit-rate is below its 0.9 aspiration and is shown as informational rather than edited to pass; title boosting raised it from 0.74 to 0.78 and the ranker is one function to replace.

Quirks and known limitations (stated, not hidden)

Offline mode proves plumbing, not prose

Its 100 % verified share is by construction; on live text it can leave a section empty (Netflix MD&A opens with tables). The LLM path composes those sections.

XBRL-derived ratios are approximations

EPS TTM sums quarterly XBRL frames and is not split-adjusted: Netflix shows P/E 8.98 after its stock split. Labelled as derived; a vendor feed would replace it in a pilot.

Unofficial price endpoints

Yahoo's chart API and Stooq CSV are keyless but unofficial; Stooq intermittently serves a bot-challenge page. Fixture mode is the demo default for that reason.

Retrieval hit-rate 0.78

Lexical ranking misses some golden queries; the KPI is informational and the ranker is swappable.

Cross-reference-sheet 10-Ks

McDonald's-style filings need a heading fallback; two sub-sections still over- or under-run slightly (backlog).

No authentication, local audit file

Single-user prototype: SSO in front of the API and a tamper-evident audit sink are pilot work, listed in the ORR.

Metrics card timestamp churn

Every gate run rewrites the rendered card's timestamp; harmless, noted for the next build.

Guard scans claim text, not quotes

Quotes are never rendered, so the exposure is nil today; noted as LOW.

Cost, performance and operability

1 call

per briefing or question; prompt capped by section character limits (≈ 27 k tokens)

33 ms

median offline briefing; live ≈ 60–90 s dominated by the model

2–3 s

cold live fetch per ticker (five documents, four facts, one chart); warm reads < 1 s

0 keys

needed for the demo or for live data; the AI gateway key is injected at runtime only

Operability

- Runbook with start/stop, modes, common failures and fallbacks
- Demo checklist with a fallback line for every failure mode
- Operational readiness review and change records for 0.1.0 and 0.2.0
- `fathom probe` checks the gateway; `fathom fetch` pre-warms live data
- Rollback is an environment variable (offline / fixture) or a git tag

Observability and audit

- Structured INFO logs: provider, model, latency, claim counts — never bodies
- Audit JSONL per call with sha256 of prompt and response
- Alert candidates named for a pilot: verified share < 0.7/day, guard hits > 0, provider error rate
- Manifest per live ticker records URLs, accessions and fetched-at for provenance

Roadmap to a pilot

Weeks 1–2

Harden data

- Vendor or licensed quote feed behind the existing price interface
- Split-adjusted fundamentals; XBRL only as fallback
- Full EDGAR universe with nightly pre-warm

Weeks 3–4

Secure and host

- SSO in front of the API and page
- Tamper-evident audit sink (hash chain or append-only store)
- Secrets from the platform vault; SBOM in CI

Weeks 5–6

Measure

- Advisor eval set built with the compliance team
- Live verified-share and guard dashboards
- Embedding or hybrid ranker if the hit-rate target is confirmed

Week 7+

Scale

- Model routing through the gateway (cost tiers)
- Batch briefings for watchlists
- Change record and ORR sign-off for production

Everything above is additive: the contracts, the verifier, the guard and the audit record do not change.

Appendix A — Decision log (abridged)

ID	Decision	Consequence
D-002	Demo data from four Hugging Face datasets, committed as fixtures with licences and anchors	Deterministic, offline demo; refresh is one script
D-003	Grounding by verbatim citation, not by trust	JSON claim contract; verifier; badges; verified-share KPI
D-004	BM25 over SEC items, no vector database	No infrastructure; explainable; one function to replace
D-005	Providers speak one JSON contract; offline is a real provider	One code path; CI and demo never need a key
D-006 / D-009	Parser heading fallback for cross-reference-sheet 10-Ks (tightened after a failed attempt)	McDonald's briefing restored; invariant for all other filings
D-007 / D-010	Claim conversion never raises; transport errors are Fathom errors	HIGH security finding and a resilience gap closed
D-008 / D-011	Title-boosted retrieval; API binds loopback; demo polish heuristics	Honest KPI; safer default; readable offline briefing
D-013	Live mode materializes the fixture schema from SEC EDGAR + Yahoo/Stooq	Any US ticker; downstream untouched; new trust boundary B6
D-014	Typed payload accessors and structural fuzz as acceptance criteria	Closed a class of parsing crashes found across four security rounds

Appendix B — Build statistics and lessons

Item	v0.1.0 (fixtures)	v0.2.0 (live data)
Task packs	18 (T-000 ... T-017)	7 (T-018 ... T-024)
Implementer attempts	22	17
Verifier runs (fresh context, pinned worktree)	20	13
Security reviews	8	9 (four rounds on payload parsing)
Real bugs caught by independent roles	5 (deps extras, fallback bodies, null completion, stale figures, bind 0.0.0.0)	6 (ticker routing, path build before validation, four parsing classes)
Ledger entries / gates	32 / G0–G8	11 / G1–G4 re-entry, G6.4, G7, G8
Tests / coverage at ship	287 / 96.0 %	458 / 95.8 %

Lessons written into the decision log

- Specs that touch unstructured text need a content-level acceptance test before dispatch (the parser fallback took two attempts).
- Documentation packs run last, once, after all code tasks (figures moved three times).
- For third-party payloads, specify typed accessors, whole-body guards and structural fuzzing up front; enumerating failure modes leaks.
- Verifiers occasionally write into the worktree copy; the orchestrator names the live path and copies before removal.

Appendix C — Repository map and how to run

fathom/	core: config, errors, data, quotes, filings, retrieval, prompts, providers, contracts, guard, audit, briefing, ask, bench, cli, api, mcp_server, live/*
app/	Streamlit page + rendering helpers
data/	fixtures (parquet) + SOURCES.md (licences, anchors)
tests/	458 tests; recorded live fixtures
metrics/	headline.json, card.json/md, render.py (drift-checked)
docs/design/	brief, requirements, HLD, threat model, LLD, plan, live-data amendment, decisions
docs/tasks/	25 packs with verdict and security files
docs/evidence/	hash-chained ledger + exported evidence pack
docs/ops/	runbook, ORR, change records, demo checklist
docs/pitch/	one-slide pitch and this deck

Run it

```
uv sync --all-extras
uv run python scripts/check.py
uv run fathom app

# live data, no keys
export FATHOM_DATA_SOURCE=live
export FATHOM_SEC_CONTACT=you@firm.com
uv run fathom fetch NFLX
uv run fathom brief NFLX

# live model via the gateway
export FATHOM_LLM_PROVIDER=portkey
export PORTKEY_API_KEY=...
uv run fathom probe
```

Questions

Fathom · roshanrana/fathom · release 0.2.0 · design docs, evidence ledger and runbook in the repository